Comp 442 Group Assignment 4

“Phish Detector”

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Revision History

Version	Date	Author(s)	Notes
v1.0	6/20	Haley	Google doc created and outline written for members
v1.1	6/21	Full team	Section 1-4 started
v1.2	6/22	Full team	Sections 5-9 started
v1.3	6/23	Haley	Full review of doc. Prepared for submission

1. Introduction

1.1 Project Overview

The Phish Detector project aims to develop a machine learning-based system capable of identifying spam and phishing messages. The project applies data preprocessing, natural language processing (NLP), and machine learning techniques to classify messages as either legitimate or malicious. By analyzing message content and extracting meaningful features, the system seeks to improve the detection of suspicious communications and support user protection against common cybersecurity threats.

The dataset used for this project consists of labeled email messages categorized as either spam or ham (legitimate). Through a combination of data preparation, feature engineering, model development, and evaluation, the project will demonstrate how machine learning can be used to address real-world spam and phishing detection challenges.

1.2 Assignment Objectives

The purpose of Assignment 4 is to prepare the dataset for future machine learning model development by applying data preprocessing techniques. Data preprocessing is an essential step because machine learning algorithms rely on clean, consistent, and well-structured data to produce accurate results.

This assignment focuses on several key preprocessing activities, including data cleaning, data transformation, feature selection, and feature extraction. Data cleaning is used to identify and address issues such as duplicate records, missing values, and outliers. Data transformation techniques, including discretization and normalization, are applied to convert the data into forms that are more suitable for analysis and modeling. Feature selection and feature extraction are used to identify and create meaningful attributes that may improve future model performance.

The overall objective of this assignment is to improve data quality, enhance data usability, and prepare a refined dataset that can be used effectively during the machine learning training and evaluation phases of the project.

2. Dataset Overview

2.1 Dataset Description

The dataset used for this project is the Email Spam Detection Dataset obtained from Kaggle. The dataset contains 5,572 labeled messages that are classified as either spam or ham (legitimate). Because the dataset already contains class labels, it is well suited for supervised machine learning and text classification tasks.

Each record in the dataset represents a single message along with its corresponding classification label. The dataset is stored in CSV format and provides a foundation for exploring message characteristics, extracting meaningful features, and developing machine learning models capable of identifying spam and phishing messages.

2.2 Dataset Characteristics

The dataset contains 5,572 records and 2 primary attributes:

- **Category** – Indicates whether a message is classified as spam or ham.
- **Message** – Contains the text content of the message.

The dataset combines structured tabular data with unstructured text data. While the overall dataset is organized into rows and columns, the Message attribute contains raw text that requires preprocessing and feature extraction before it can be used by machine learning algorithms.

The dataset also exhibits class imbalance, with ham messages significantly outnumbering spam messages. This characteristic will be considered during model development and evaluation.

2.3 Initial Data Quality Assessment

An initial assessment of the dataset was conducted prior to preprocessing. The assessment focused on identifying potential issues that could negatively affect machine learning performance, including missing values, duplicate records, inconsistent formatting, and outliers.

The preliminary analysis revealed no missing values within the dataset. However, duplicate records and several outliers were identified and investigated further during the data cleaning phase. The assessment also confirmed that the dataset contains sufficient information for feature extraction and future machine learning model development.

3. Data Cleaning

3.1 Missing Value Analysis

A missing value analysis was performed to determine whether the dataset contained incomplete or null records. Missing values can negatively impact machine learning performance by reducing data quality and introducing inconsistencies during model training.

The analysis showed that no missing values were present in either the Category or Message attributes. Because the dataset contains complete records, no imputation or missing value treatment was required.

3.2 Duplicate Record Detection

Duplicate records were analyzed to identify repeated messages that could potentially bias future machine learning models. Duplicate messages may cause certain patterns to be overrepresented during training, resulting in reduced generalization performance.

The duplicate detection process identified 415 duplicate records within the dataset. These records represented repeated message content and were considered unnecessary for model development.

3.3 Duplicate Record Removal

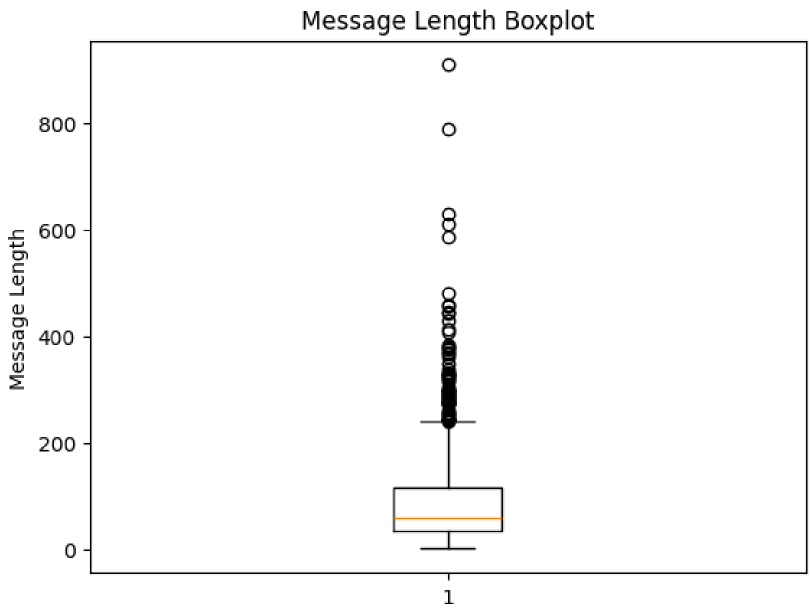
After identifying duplicate records, duplicate messages were removed from the dataset. Removing duplicates helps reduce bias, improve dataset quality, and ensure that machine learning models learn from unique examples rather than repeated observations.

Following duplicate removal, the dataset size decreased from 5,572 records to 5,157 records. The cleaned dataset provides a more reliable foundation for future feature engineering and model training activities.

3.4 Outlier Detection

Outlier detection was performed using statistical visualization techniques, including boxplots of message length. Outliers represent observations that differ substantially from the majority of the dataset and may influence statistical measures or machine learning performance.

The analysis revealed several unusually long messages compared to the typical message length distribution. These outliers contribute to the positive skewness observed during statistical analysis. While the outliers were identified, they were retained because they may represent legitimate examples of message variation that could provide useful information during model training.



3.5 Data Cleaning Summary

The data cleaning process confirmed that the dataset contains no missing values, identified 415 duplicate records, and detected several outliers in message length. Duplicate records were removed to improve data quality, while outliers were retained for further analysis. Overall, the data cleaning process improved the consistency and reliability of the dataset and prepared it for the transformation and feature engineering stages of preprocessing.

4. Data Transformation

4.1 Discretization

Discretization was applied to the Message Length feature to convert continuous numerical values into categorical groups. This process helps simplify analysis and allows patterns to be examined using broader categories rather than individual numerical values.

Message lengths were grouped into three categories:

- **Short** – Messages with relatively low character counts
- **Medium** – Messages with moderate character counts

- **Long** – Messages with high character counts

By grouping message lengths into categories, the dataset becomes easier to interpret and analyze. These categories may also be useful for future exploratory analysis and feature engineering.

4.2 Normalization

Normalization was performed on the Message Length feature using Min-Max Scaling. Normalization transforms values to a common scale, typically between 0 and 1, while preserving the relative relationships between observations.

Machine learning algorithms often perform better when numerical features are scaled consistently. Without normalization, features with larger numerical ranges may have a greater influence on model behavior than intended.

The normalized Message Length feature provides a standardized representation of message size that can be used alongside other engineered features during future model training.

4.3 Transformation Results

The transformation process successfully converted the Message Length feature into both categorical and normalized representations. Discretization created the Length Category feature, while normalization generated a scaled version of Message Length.

These transformations improve data usability and provide additional representations of the original feature that may support future machine learning models. The transformed features also help ensure that the dataset is better prepared for subsequent feature engineering and classification tasks.

5. Feature Selection

5.1 Selected Features

Several features were selected from the dataset for future machine learning model development. These features were chosen because they capture characteristics that may help distinguish spam messages from legitimate messages.

The selected features include:

- Message Length
- Word Count
- Digit Count
- URL Count
- Length Category
- Normalized Message Length

These features were either extracted directly from the message content or generated through preprocessing and transformation techniques.

5.2 Feature Selection Rationale

Feature selection is important because it helps reduce complexity, improve model interpretability, and focus machine learning algorithms on the most informative attributes. Rather than using all available information, the project focuses on features that are likely to contribute meaningful predictive value.

Message Length and Word Count provide information about the overall size and structure of a message. Digit Count may help identify messages containing excessive numerical content, which is commonly observed in spam messages. URL Count is useful because spam and phishing messages frequently contain links that encourage users to visit external websites. Length Category and Normalized Message Length provide alternative representations of message size that may improve model performance.

By selecting these features, the project reduces unnecessary complexity while preserving information that may be useful for future spam classification tasks.

6. Feature Extraction

6.1 Message Length

Message Length was extracted by calculating the total number of characters contained within each message. This feature provides information about the overall size of a message and may help identify patterns associated with spam messages, which often differ in length from legitimate messages.

The Message Length feature was also used for statistical analysis, visualization, normalization, and discretization during earlier phases of the project.

6.2 Word Count

Word Count was extracted by counting the number of words contained in each message. This feature provides insight into message complexity and content density. Messages containing more words generally have greater overall length and may exhibit different characteristics depending on whether they are spam or legitimate.

Word Count also demonstrated a strong positive correlation with Message Length during exploratory data analysis.

6.3 Digit Count

Digit Count was extracted by counting the number of numerical characters present within each message. Spam messages frequently contain phone numbers, promotional codes, prices, or other numerical information, making this feature potentially useful for classification.

By quantifying the amount of numerical content, this feature provides additional information that may help distinguish spam messages from legitimate communications.

6.4 URL Count

URL Count was extracted by identifying and counting hyperlinks or web addresses contained within each message. Spam and phishing messages often encourage users to visit external websites, making the presence of URLs an important indicator of suspicious activity.

This feature may contribute valuable information during model training by helping identify messages that contain potentially risky links.

6.5 Feature Extraction Summary

The feature extraction process transformed raw text messages into structured numerical attributes that can be used by machine learning algorithms. The extracted features include Message Length, Word Count, Digit Count, and URL Count, along with transformed versions such as Length Category and Normalized Message Length.

These features provide measurable representations of message characteristics and will serve as inputs for future machine learning models developed during later stages of the project.

7. Data Integration

7.1 Data Integration Assessment

Data integration was evaluated as part of the preprocessing process to determine whether information from multiple datasets or external sources needed to be combined. Data integration is commonly used when relevant information is distributed across multiple files, databases, or data sources.

For this project, data integration was not required because all necessary information was contained within a single dataset. The selected Email Spam Detection Dataset already included both the message content and the corresponding classification labels needed for analysis and future model development.

As a result, no dataset merging, schema alignment, or conflict resolution procedures were necessary. The project proceeded using a single, unified dataset throughout the preprocessing process.

8. Findings and Observations

8.1 Data Quality Findings

The data quality assessment revealed that the dataset contained no missing values, indicating that all records were complete and usable for analysis. However, 415 duplicate records were identified and removed during the data cleaning process. The dataset also contained several outliers, particularly within the Message Length feature, which contributed to the observed positive skewness in the data.

8.2 Transformation Findings

Data transformation techniques improved the consistency and usability of the dataset. Message Length was successfully discretized into categorical groups and normalized using Min-Max Scaling. These transformations produced additional feature representations that may improve model performance and support future analysis.

8.3 Feature Engineering Findings

Feature extraction produced several meaningful numerical attributes, including Message Length, Word Count, Digit Count, and URL Count. These features provide measurable representations of message characteristics that may help distinguish spam messages from legitimate messages. The resulting feature set establishes a strong foundation for future machine learning model development and evaluation.

9. Conclusion

This assignment focused on preparing the Email Spam Detection Dataset for future machine learning development through data preprocessing techniques. Data cleaning activities identified and removed duplicate records, while missing value analysis confirmed that the dataset contained complete records. Data transformation techniques such as discretization and normalization improved data consistency and usability, and feature extraction produced several meaningful attributes that can support spam classification.

Overall, the preprocessing process improved the quality of the dataset and established a structured set of features suitable for machine learning applications. The resulting dataset is now better prepared for model training, evaluation, and deployment during the next stages of the Phish Detector project.

Appendices

No calls were needed to complete this assignment as we collectively worked on it through the doc and split the sections.